

PfSense Assignment

LAB1 - Networking & Interface Configuration

Interfaces / WAN (em1)

General Configuration

Enable

☒ Enable interface

Description

WAN

Enter a description (name) for the interface here.

IPv4 Configuration Type

DHCP

IPv6 Configuration Type

DHCP6

MAC Address

xxxxxxxxxxxx

This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xxxxxxxxxx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex

Default (no preference, typically autoselect)

Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

DHCP Client Configuration

Options

☐ Advanced Configuration

☐ Configuration Override

WARNING:

The password for this account is insecure. Password is currently set to the default value (pfSense).
Change the password as soon as possible.

Interfaces / WAN (em1)

The WAN configuration has been changed.
The changes must be applied to take effect.
Don't forget to adjust the DHCP Server range if needed after applying.

☒ Apply Changes

General Configuration

Enable

☒ Enable interface

Description

WAN

Enter a description (name) for the interface here.

IPv4 Configuration Type

Static IPv4

IPv6 Configuration Type

DHCP6

MAC Address

xxxxxxxxxxxx

This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xxxxxxxxxx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address	192.168.10.1	/ 24
IPv4 Upstream gateway	None	+ Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a **WAN type interface**.
Gateways can be managed by [clicking here](#).

DHCPv6 Client Configuration

Options	☐ Advanced Configuration Use advanced DHCPv6 configuration options.	☐ Configuration Override Override the configuration from this file.
Use IPv4 connectivity as parent interface	☐ Request a IPv6 prefix/information through the IPv4 connectivity link	
Request only an IPv6 prefix	☐ Only request an IPv6 prefix, do not request an IPv6 address	
DHCPv6 Prefix Delegation size	64 The value in this field is the delegated prefix length provided by the DHCPv6 server. Normally specified by the ISP.	
Send IPv6 prefix hint	☐ Send an IPv6 prefix hint to indicate the desired prefix size for delegation	
Do not wait for a RA	☐ Required by some ISPs, especially those not using PPPoE	

Reserved Networks

Block private networks and loopback addresses	☐
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Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127.0). This option should generally be enabled, unless this network interface resides in such a

The WAN configuration has been changed.
The changes must be applied to take effect.
Don't forget to adjust the DHCP Server range if needed after applying.

[Apply Changes](#)

General Configuration

Enable	☒ Enable interface
Description	WAN Enter a description (name) for the interface here.
IPv4 Configuration Type	Static IPv4
IPv6 Configuration Type	DHCP6
MAC Address	xxxxxxxxxxxx This field can be used to modify ("spoof") the MAC address of this interface. Enter a MAC address in the following format: xxxxxxxx:xxxx:xx or leave blank.
MTU	 If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS	 If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.
Speed and Duplex	Default (no preference, typically autoselect) Explicitly set speed and duplex mode for this interface. WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

```

0) Shell

Enter an option: 1

Valid interfaces are:

em0      08:00:27:11:c4:f3   (up) Intel(R) Legacy P80/1000 MT 82540EM
em1      08:00:27:16:2d:34   (up) Intel(R) Legacy P80/1000 MT 82540EM

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now (y/n)? y

VLAN Capable interfaces:

em0      08:00:27:11:c4:f3   (up)
em1      08:00:27:16:2d:34   (up)

Enter the parent interface name for the new VLAN (or nothing if finished): em0
Enter the VLAN tag (1-4094): WAN Uplink

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7) Ping host          16) Restart PHP-FPM
0) Shell

Enter an option:

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

eVirtualBox Virtual Machine - Netgate Device ID: 3e92c1ed25e58a4e1fcf

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.56.102/24
LAN (lan) -> em1 -> v4/DHCP4: 192.168.56.101/24

0) Logout / Disconnect SSH      9) pftop
1) Assign interfaces            10) Filter Logs
2) Set interface(s) IP address  11) Restart GUI
3) Reset admin account and password 12) PHP shell - pfSense tools
4) Reset to factory defaults     13) Update from console
5) Reboot system                14) Enable Secure Shell (sshd)
6) Halt system                  15) Restore recent configuration
7) Ping host                    16) Restart PHP-FPM
8) Shell

Enter an option:

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LAB2-Firewall Rule Logic & Policy Enforcement

isense

COMMUNITY EDITION

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

Go

Status / System Logs / Firewall / Normal View

System

Firewall

DHCP

Authentication

IPsec

PPP

PPPoE/L2TP Server

OpenVPN

NTP

Packages

Settings

Normal View

Dynamic View

Summary View

La: Rule details

Action: block

Reason: ip-option

Tracker ID: 1757202313

Matched Rule: unavailable

Associated Rules:

@70 pass in quick on vmx1 inet proto igmp from <LAN__NETWORK:1> to 239.255.255.250 keep state (if-bound) label "USER_RULE: Passed via EasyRule" label "id:1757202313" identifier 1757202313

Sep 7 10:17:10

LAN

Passed via EasyRule (1757202313)

192.168.1.45

239.255.255.250

IGMP

Sep 7 10:17:05

LAN

Passed via EasyRule (1757202313)

192.168.1.45

239.255.255.250

IGMP

Sep 7 10:17:05

LAN

Passed via EasyRule (1757202313)

192.168.1.70

239.255.255.250

IGMP

Sep 7 10:16:03

LAN

Passed via EasyRule (1757202313)

192.168.1.155

239.255.255.250

IGMP

Sep 7 10:16:02

LAN

Passed via EasyRule (1757202313)

192.168.1.45

239.255.255.250

IGMP

Sep 7 10:15:07

LAN

Passed via EasyRule (1757202313)

192.168.1.45

239.255.255.250

IGMP

Sep 7 10:15:06

LAN

Passed via EasyRule (1757202313)

192.168.1.155

239.255.255.250

IGMP

Sep 7 10:15:01

LAN

Passed via EasyRule (1757202313)

192.168.1.45

239.255.255.250

IGMP

17 matched log entries.Max(50)					
Act	Time	If	Source	Destination	Proto
	Oct 19 01:51:46	LAN	192.168.1.10	159.153.226.105	ICMP
	Oct 19 01:51:45	LAN	192.168.1.10	159.153.225.30	ICMP
	Oct 19 01:51:43	LAN	192.168.1.10	159.153.93.2	ICMP
	Oct 19 01:51:42	LAN	192.168.1.10	64.125.199.186	ICMP
	Oct 19 01:51:40	LAN	192.168.1.10	64.125.31.206	ICMP
	Oct 19 01:51:39	LAN	192.168.1.10	64.125.25.113	ICMP
	Oct 19 01:51:37	LAN	192.168.1.10	64.125.25.46	ICMP
	Oct 19 01:51:36	LAN	192.168.1.10	64.125.30.233	ICMP
	Oct 19 01:51:34	LAN	192.168.1.10	64.125.31.234	ICMP
	Oct 19 01:51:33	LAN	192.168.1.10	64.125.24.5	ICMP
	Oct 19 01:51:32	LAN	192.168.1.10	75.149.228.134	ICMP
	Oct 19 01:51:30	LAN	192.168.1.10	68.86.87.18	ICMP
	Oct 19 01:51:29	LAN	192.168.1.10	68.86.91.229	ICMP
	Oct 19 01:51:27	LAN	192.168.1.10	68.85.155.14	ICMP
	Oct 19 01:51:26	LAN	192.168.1.10	68.85.154.10	ICMP
	Oct 19 01:51:24	LAN	192.168.1.10	162.151.1.141	ICMP
	Oct 19 01:51:23	LAN	192.168.1.10	67.160.236.1	ICMP

pfSense

COMMUNITY EDITION

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Status / System Logs / Firewall / Normal View

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Summary View

La: Rule details

Action: block

Reason: ip-option

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×	Sep 7 10:17:10	LAN	Passed via EasyRule (1757202313)	192.168.1.45	239.255.255.250	IGMP
×	Sep 7 10:17:05	LAN	Passed via EasyRule (1757202313)	192.168.1.45	239.255.255.250	IGMP
×	Sep 7 10:17:05	LAN	Passed via EasyRule (1757202313)	192.168.1.70	239.255.255.250	IGMP
×	Sep 7 10:16:03	LAN	Passed via EasyRule (1757202313)	192.168.1.155	239.255.255.250	IGMP
×	Sep 7 10:16:02	LAN	Passed via EasyRule (1757202313)	192.168.1.45	239.255.255.250	IGMP
×	Sep 7 10:15:07	LAN	Passed via EasyRule (1757202313)	192.168.1.45	239.255.255.250	IGMP
×	Sep 7 10:15:06	LAN	Passed via EasyRule (1757202313)	192.168.1.155	239.255.255.250	IGMP
×	Sep 7 10:15:01	LAN	Passed via EasyRule (1757202313)	192.168.1.45	239.255.255.250	IGMP

```
4) Reboot system
5) Reboot system
6) Halt system
7) Ping host
8) Shell

13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: ipconfig em1

VirtualBox Virtual Machine - Netgate Device ID: 8d1c40dc0e30eb303762
*** Welcome to pfSense 2.0.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em1 -> 04/BHCP4: 192.168.56.101/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reboot system
4) Reboot system
5) Reboot system
6) Halt system
7) Ping host
8) Shell

9) pTop
10) Filter Logs
11) Restart GUI
12) PHP shell - pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: 
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